

Experiment No. 04

Aim: Verification of Ohm's law

APPARATUS REQUIRED: 30V dc supply, Digital Multimeter as an Ammeter, Digital Multimeter as a Voltmeter, Resistance.

THEORY: Multimeter is common multipurpose instrument used to measure electrical quantities in a circuit. A Multimeter used to check AC or DC supply, Resistances, small amount of current, continuity and various electrical components. There are major differences between the analog and digital multimeter. Digital Multimeter offers very automatic range and easy to read display.

If I is the current flowing through a conductor of resistance R across which a potential difference V is applied then according to ohm's law

$$I \propto V \text{ or } I = V/R$$

Where V in volts, R is in ohms, I is in amperes and $1/R$ is the coefficient of proportionality.

As long as physical states i.e. temperature, pressure etc. remain the same the current flowing through a conductor is directly proportional to the potential difference applied across its ends.

PROCEDURE:

- 1) Fix the value of R as 100 ohms.
- 2) By varying R_h voltage across R will change, record voltmeter and ammeter readings.
- 3) Take at least 5 readings. Calculate the ratio of V and I for each reading and tabulate as in the observation table.
- 4) Draw graphs of I versus V for different value of Resistance.
- 5) Comment about the nature of the graphs drawn by you.

DIAGRAM:

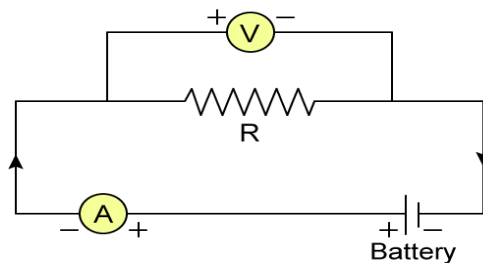


Figure: Circuit diagram for Ohm's Law verification

OBSERVATION TABLE:

S.No	Voltage (V) volts	Current (I) Amp	Calculate value of $R=(V/I)$ Ohm
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

CONCLUSION AND RESULT:

The ratio of V and I for any set of readings is constant and is equal to R. This relationship, $(V/I) = R$,

or $V=IR$ was first established by a scientist ohm and is Known as Ohm's Law.

You will observe that the nature of the graphs is straight lines passing through the origin having different slopes. For higher values of R, the slopes of the graphs will move towards x – axis

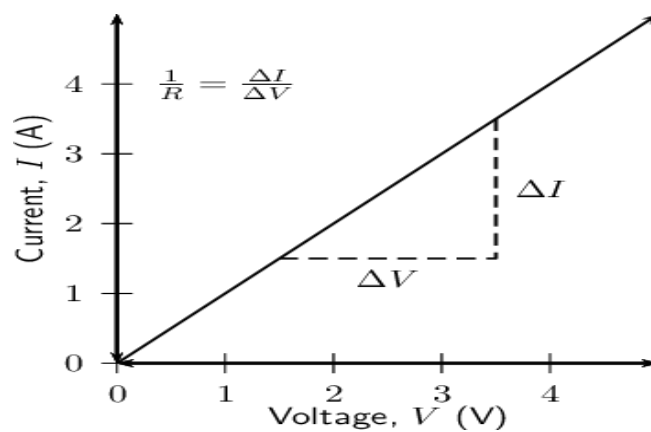


Figure: Graph between voltage and current

PRECAUTIONS:

- 1) Switch off the supply first and then start making connections.
- 2) Connections should be neat and tight.
- 3) Do not touch the bare conductors or terminals without switching off the supply.
- 4) Adjust the zero error of the instrument.
- 5) Meters of suitable range should be used.
- 6) After the completion of experiment switch off the supply.

VIVA:

1. State Ohm's Law?
2. What does $I \propto V$ and $V \propto I$ represent?
3. What do mean by resistance?
4. If the value of the resistance provided by the conductor is increased, then the slope of I versus V Curve will shift towards which axes?
5. Is the Ohm's Law is applicable for capacitor or inductor circuit?
6. What will happen to the current if we stretch the conductor to the double of its length?
7. If the surrounding temperature is increased, then what will happen to the resistance of the conductor made of copper?